MIL-P-3631C
14 October 1981
SUPERSEDING
MIL-P-3631B
17 June 1976

MILITARY SPECIFICATION

PARALLEL RULERS, (ROLLING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rolling parallel rulers.

1.2 Classification. Rulers covered by this specification shall be the following sizes, as specified (see 6.2):

Size A - 12 inches long.
Size B - 18 inches long.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-16T20:33Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 276 - Stainless and Heat-Resisting Steel Bars and Shapes.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The rulers shall consist of a base with two parallel, beveled drawing edges, and a roller assembly to permit parallel movement of the rulers.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.2.1 Corrosion-resisting steel. Corrosion-resisting steel shall conform to ANSI/ASTM A 276, UNS S20200 or S30200, Condition A.

3.3 Base. The base shall be corrosion-resistant steel. The width and thickness of the base shall conform to present commercial practice. The length of the rulers shall be 12 or 18 inches (see 1.2), plus or minus 0.125 inch.

3.3.1 Straightness. Each drawing edge of the ruler shall be beveled and shall be straight within 0.006 inch.

3.3.2 Parallelism. The drawing edges of the ruler shall be parallel within 0.01 inch per foot.

3.4 Roller assembly. The roller assembly shall consist of an axle, two rollers, and two bearing standards, and shall be the same material as the base. The axle shall be suspended between standards mounted as near as practicable to the ends of the base. The axle shall be free to rotate and shall have no end or lateral play. A roller shall be rigidly attached near each end of the axle and shall turn with the axle. The diameter of the rollers shall be such so as to protrude slightly below the underside of the base. The rolling surface of each roller shall be knurled, scored, or grooved to prevent slipping when the ruler is guided over a drafting medium. The axis of the roller assembly shall be parallel with the drawing edges. The rollers shall be such size and alinement so that parallel motion will be obtained within 0.025 inch per foot of length of the drawing edge.

3.5 Axle cover. The axle cover shall be the same material as the base and shall be shaped to provide a convenient handle for lifting or moving the ruler.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

3.6 Finish. Unless otherwise specified (see 6.2), the ruler shall be finished in accordance with the contractor's standard practice.

3.7 Identification marking. Each ruler shall be permanently identified in accordance with MIL-STD-130.

3.8 Carrying case. When specified (see 6.2), an individual carrying case shall be furnished; construction shall be at the contractor's option.

3.9 Workmanship. The rulers shall be free from defects such as nicks, scratches, and any other defect which will impair their serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications, and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

4.4 Inspection procedure.

4.4.1 Examination. The rulers shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Length not within the tolerances specified.
103. Roller assembly binds, or does not rotate freely.
104. End play or sideplay in the roller assembly.
105. Rollers not knurled, scored, or grooved.
106. Axle cover not as specified.
107. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.4.2 Tests.

4.4.2.1 Drawing edge straightness. The drawing edge shall be placed against a precision flat, which shall be flat to within 0.001 inch total. The separation between the drawing edge and the precision flat shall be measured at intervals not to exceed 3 inches with a thickness gage which has an accuracy of not less than 0.001 inch. Repeat this test using the other drawing edge. Nonconformance to 3.3.1 shall constitute failure of this test.

4.4.2.2 Parallelism. The distance between the two drawing edges shall be measured with a micrometer. Nonconformance to 3.3.2 shall constitute failure of this test.

4.4.2.3 Parallel rolling. The ruler shall be placed on a suitable drawing medium and a fine line drawn along the length of one drawing edge. The ruler shall then be moved on its rollers a distance of 12 inches, and another fine line drawn the entire length of the same drawing edge. Perpendicular lines shall then be drawn within 0.50-inch from the ends of the first line to the second line and the distance between the first two lines measured along the perpendicular lines. Nonconformance to 3.4 shall constitute failure of this test.

4.4.2.4 Carrying case drop test. The case shall be dropped from a height of 48 inches onto a floor comprising of not more than 2-inch-thick plywood backed by concrete. The case shall be dropped four times: twice on different faces and twice on different corners. The first case to be drop tested shall have a ruler inside during the test. After the drop test is completed, the ruler shall be examined in accordance with 4.4.1 and then tested in accordance with 4.4.2. Nonconformance to 4.4.1 and 4.4.2 will constitute failure of this test. In all subsequent drop tests, a dummy load of equal mass and form to a ruler shall be used and put inside the case to be drop tested. The case shall be observed during the test for opening, thus providing no protection; after the test, the case shall be inspected. Evidence of damage that would prohibit the case from protecting the ruler shall constitute a failure.

Source: https://assist.dla.mil -- Downloaded: 20261O-16T20:33Z
Check the source to verify that this is the current version before use.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, or containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.
109.	Consolidation not as specified for Level A.
110.	Quantities packed together exceed that specified for Level A or Level B.
111.	Enclosure not as specified for Level B.
112.	Preservation and packing not in accordance with the referenced document as specified for Commercial.
113.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each ruler, contained in the case when furnished, shall be preserved in accordance with MIL-P-116, Method IC-2 using the contractor's standard commercial box or a close-fitting box conforming to PPP-B-566.

5.1.1.1 Consolidation. Rulers of like description shall be consolidated together in quantities not to exceed the weight limitation of the box in close-fitting boxes conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. Rulers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Rulers of like description, preserved as specified in 5.1 shall be packed together in close fitting boxes conforming to PPP-B-601, Overseas type, style optional, in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Rulers of like description, preserved as specified in 5.1, shall be packed together in quantities not to exceed the size and weight limitation of the box, in close-fitting boxes conforming to PPP-B-636, V3c, style optional. Box closure and sealing shall be as specified for Method V in the Appendix to the box specification.

5.2.3 Commercial. Rulers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The rulers are intended for use in drawing parallel lines in general drafting practice.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size ruler required (see 1.2).
(c) When the finish shall be other than as specified (see 3.6).
(d) When carrying cases are required (see 3.8).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.4 Classification change. Classification has been added as follows:

Size A - 12 inches long

Size B - 18 inches long

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-0712

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-3631C Parallel Rulers, (Rolling)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-16T20:33Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T20:33Z
Check the source to verify that this is the current version before use.